

Reflective Journal: Module 05 Lab - Data Preparation

Module 05 helped me understand that machine learning is not just about choosing a model and running it. A big part of the process happens before the model ever gets trained. In Module 04, we focused more on exploring the data and finding patterns, but this lab moved into the next step: actually preparing the data so a machine learning model can use it. That made the whole workflow feel more realistic to me because raw data is usually messy, incomplete, and not automatically ready for modeling.

The Titanic dataset was a good example for this lab because it had common problems that show up in real datasets. One of the first issues was missing values, especially in the Age column. Instead of just deleting rows with missing ages, we used median imputation to fill them in. I thought this made sense because deleting too many rows could remove useful information, and the median is safer than the mean when there are outliers or skewed values. This helped me see that cleaning data is not just a technical step. It also involves making choices and understanding why one method may be better than another.

Another important part of the lab was encoding categorical data. Before this module, I understood that models need numbers, but this lab made that idea more practical. Columns like Sex and Embarked are meaningful to humans, but a machine learning model cannot directly understand words like "male," "female," or "S." One-hot encoding fixed that by turning categories into separate numeric columns with 0s and 1s. This was one of the clearer examples of data preparation for me because it showed how we translate real-world information into something a model can actually process.

The scaling part of the lab also helped me understand why feature size matters. Age and Fare are both numerical columns, but they exist on different scales. A fare value can be much larger than an age value, and certain models might accidentally treat the larger numbers as more important just because of their size. Using StandardScaler made the values more comparable by centering them around a common scale. I also liked learning that scaling is not equally important for every model. For example, Decision Trees usually do not need scaling because they split data based on conditions rather than distance.

The key readings and course presentation also helped me connect this lab to the bigger picture of feature engineering. One idea that stood out to me is that feature engineering is about improving how the data represents the actual problem. Good features can help a simple model perform better, while on the other hand messy or poorly chosen features can hurt even a strong model. That made me think differently about machine

learning because the “smart” part is not only the algorithm. A lot of the intelligence comes from how carefully the data is prepared.

One challenge in this lab was making sure the notebook structure and code were clean before running everything. Like the previous lab, some of the code needed to be placed carefully so each command ran on its own line. This reminded me that small formatting or syntax issues can stop the whole notebook, even when the overall logic is correct. I am starting to get more comfortable debugging those issues, but it still takes patience.

Overall, this lab made data preparation feel more important and less like a background step. I learned how missing values, categorical variables, and scaling can all affect whether a dataset is usable for machine learning. The biggest takeaway for me is that a model can only learn from the data it is given. If the data is incomplete, messy, or in the wrong format, the model’s results will probably suffer. Going forward, I think I will pay more attention to the preparation stage before jumping into model training because this lab showed me that clean, thoughtful data is one of the foundations of good machine learning.